

Cross-Lingual NER Transfer from English to Danish using BERT-based Models

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Abstract

This project studies how to improve Danish Named Entity Recognition (NER) by transferring knowledge from English with BERT-based models. Because Danish has less labeled data, we compare zero-shot transfer, transfer plus Danish fine-tuning, and a Danish-only baseline. We use mBERT and XLM-R for transfer and include an English BERT baseline. We also test whether English and Danish being related Indo-European languages gives a practical transfer advantage.

1 Introduction and Background

NER needs labeled data, which is abundant for English but limited for Danish. This makes Danish NER harder to train with only supervised data. A practical solution is transfer learning: train on English first, then adapt to Danish.

Earlier work supports this idea. Plank (2020) reported English-to-Danish transfer with zero-shot F1 around 58–61%, and above 70% when small Danish labeled sets were added. Fu et al. (2023) found transfer is usually stronger for related languages, and Sohn et al. (2024) showed that surface similarity also helps. Since English and Danish are related Indo-European languages, this pair is a strong test case.

2 Research Questions

Our main question is: *How much can English-to-Danish transfer improve Danish NER with multilingual BERT-based models?* We focus on three sub-questions:

- How well do simple monolingual English and Danish BERT baselines perform?
- How much performance do we lose in zero-shot English-to-Danish transfer compared with supervised Danish training?

- Is two-step training (first English, then Danish) better than both zero-shot transfer and Danish-only training?

We also test whether English and Danish being related languages gives a measurable transfer advantage, in line with the findings of Fu et al. (2023).

3 Methodology

We use token classification with transformer encoders in four setups: (1) an English baseline with `bert-base-cased`, (2) zero-shot transfer to Danish with mBERT and XLM-R trained on English, (3) transfer fine-tuning where those models are further trained on Danish, and (4) a Danish-only baseline. This design lets us separate three effects: pure transfer, transfer plus target-language adaptation, and fully local Danish training.

To study data efficiency, we also fine-tune with 10%, 25%, and 50% of Danish training data.

We report entity-level F1, precision, and recall, and we include per-label and error analysis (boundary mistakes, label confusion, and tokenization issues). For reproducibility, we keep random seeds fixed and log all hyperparameters.

4 Expected Outcome

We expect zero-shot transfer to be below fully supervised Danish training but above a weak baseline. We expect transfer fine-tuning to give the best Danish results, especially with limited Danish data.

References

- Yingwen Fu, Nankai Lin, Boyu Chen, Ziyu Yang, and Shengyi Jiang. 2023. [Cross-lingual named entity recognition for heterogeneous languages](#). *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 31:371–382.
- Barbara Plank. 2020. [Neural cross-lingual transfer and limited annotated data for named entity recognition](#)

[in danish](#). In *Proceedings of the 23rd Nordic Conference on Computational Linguistics (NoDaLiDa)*, pages 344–349.

Jimin Sohn, Haeji Jung, Alex Cheng, Joeeon Kang, Yilin Du, and David R. Mortensen. 2024. [Zero-shot cross-lingual ner using phonemic representations for low-resource languages](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.